

# Weiyou Li

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 Weiyou Li

 BMath, Pure Mathematics, Statistics, [University of Waterloo](#), Expected 2027

## Summary of Qualifications

- **Machine Learning:** Hands-on experience implementing **ResNet-based CNNs**, ANNs, and Logistic Regression using **PyTorch**, **TensorFlow**, and **Scikit-learn** for medical and predictive modeling.
- **Mathematical Statistics:** Strong foundation in Pure Mathematics; proficient in **stochastic modeling**, longitudinal data analysis, polynomial fitting, and Bayesian inference.
- **Data Engineering:** Expert in building automated **Python** pipelines for Cloud-to-local synchronization, large-scale medical image preprocessing (Segmentation, QC), and dataset curation.
- **Technical Stack:** Advanced Python, R, Matlab, C, SQL, Git, and  $\text{\LaTeX}$ .

## Work Experience

### Research Assistant

Unity Health Toronto — TUPIL, PI: [Dr. Eno Hysi](#)

Toronto, ON

Jan 2026 – Present

- Built and optimized **ResNet-based CNN models** for kidney fibrosis classification by applying transfer learning and cross-validation across multiple ultrasound input configurations, enabling comparison of learned image features against clinical benchmarks on 3,200+ images.
- Modeled longitudinal **eGFR trajectories** for 150+ patients by using rule-based filtering, temporal alignment, GAMM, and polynomial fitting on raw clinical Excel data, extracting baseline kidney function and slope-based features to support future kidney-function prediction.
- Developed automated **Python data-ingestion pipelines** to transfer ultrasound data from cloud platforms to local storage, reducing ingestion time by approximately 70%, eliminating manual download errors, and maintaining 100% naming-convention accuracy across 730+ patients.

### Research Assistant

University of Waterloo

Waterloo, ON

Sept 2025 – Present

- **Designed and optimized mathematical algorithms** in Python to validate theoretical conjectures, resolving algorithmic bottlenecks and reducing computational overhead by **17%** for large-scale statistical simulations.
- **Developed rigorous validation protocols** for algorithmic performance, ensuring that theoretical bounds were met across diverse data distributions and edge cases.

### Statistics Tutor

Humber Polytechnic

Etobicoke, ON

May 2025 – Aug 2025, Sept 2024 – Dec 2024

- Applied **statistical modeling**, **time-series analysis**, and debugging support to improve student learning, increasing engagement by 17%, improving performance by 15%, and achieving a **4.9/5 satisfaction** rating across 100+ sessions.

## Research & Technical Projects

### Graduation Admission Prediction (*PyTorch, TensorFlow, Scikit-learn*)

[Graduate-Admission](#)

*Supervised by Yuan Wu (Duke University)*

June 2025 - Sept 2025

- Developed and compared **Artificial Neural Network (ANN)** and **Logistic Regression** models on a 400 sample dataset to **optimize algorithm selection** for low-data scenarios.
- Achieved a 6% increase in AUC and a 10% increase in accuracy by conducting a **hyperparameter optimization** study focused on hidden layer dimensionality, epoch count, and batch sizing.
- Applied **Pearson correlation analysis** to identify and utilize the top 5 predictive features, significantly **enhancing the interpretability** of the final Logistic Regression model.

### Beyond VaR: ETF Drawdown Recovery & Survival Risk Modeling

[Beyond VaR](#)

(*Python, Pandas, scikit-learn, lifelines, Matplotlib, Financial Backtesting*)

April 2026 - Present

- Built an end-to-end **quantitative risk framework** across multiple ETFs to model drawdowns, VaR breaches, and **time-to-recovery risk** beyond traditional VaR to support portfolio allocation decisions and demonstrate ML performance in low-data environments.
- Engineered return, volatility, drawdown, VaR-breach, and recovery-event features, applied **Kaplan–Meier curves**, **Cox models**, and **group comparisons** across risk regimes.
- Combined survival-risk scores with statistical and machine learning return predictions to compare **baseline**, **risk-filtered**, and **volatility-targeted long/cash strategies**.
- Evaluated performance against buy-and-hold using **Sharpe**, **Sortino**, **maximum drawdown**, **VaR**, **Expected Short-fall**, and **recovery-time metrics**.